

IBM Applied Data Science Capstone

Will the Falcon 9 first stage land?

Data Science Capstone Project Report — predicting SpaceX booster landings to estimate launch cost



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GitHub repository: <https://github.com/Mostafa-H-Sawan/IBM-Data-Science-Capstone-SpaceX.git>

Contents: Executive Summary · Introduction · Methodology · EDA (visualization & SQL) · Folium maps · Plotly Dash · Classification results · Innovative insights · Conclusion

Executive Summary

Methodology

- Collected Falcon 9 launch data from the SpaceX REST API and by scraping Wikipedia
- Cleaned the data, filled missing payload mass, and turned 8 landing outcomes into a 0/1 label
- Explored patterns with pandas, seaborn and 10 SQL queries (SQLite)
- Mapped launch sites and their surroundings with Folium; built an interactive Plotly Dash dashboard
- Trained and tuned Logistic Regression, SVM, Decision Tree and KNN with 10-fold GridSearchCV

Results

67%

of 90 Falcon 9 launches landed the first stage — 86% in 2019-2020

77%

landing success at KSC LC-39A, the best launch site

83.3%

test accuracy for all four tuned models; Decision Tree best in validation (87.5%)

27% → 87%

landing rate of first-flight cores vs. reused cores

Insight Landing success is driven mostly by experience and hardware maturity (flight number, booster block, reuse), not by the launch site alone.

Introduction

Background and problem

SpaceX advertises Falcon 9 launches at about \$62 million, while other providers charge upward of \$165 million.

Most of the savings come from recovering and reusing the first stage. If we can predict whether the first stage will land, we can estimate the cost of a launch.

That estimate is valuable to a competitor ("Space Y") that wants to bid against SpaceX for a launch contract.

Questions we answer

- Q1** How do payload mass, orbit, launch site, flight number and booster version affect landing success?
- Q2** Does the success rate improve over time?
- Q3** What do the launch sites have in common geographically?
- Q4** Can a machine learning model predict the landing outcome, and which algorithm works best?

Insight The target variable is Class: 1 if the first stage landed, 0 if it did not.

Data collection methodology

SpaceX REST API

- 1 `requests.get()` on the `launches/past` endpoint (static JSON)
- 2 `pd.json_normalize()` → DataFrame; keep rocket, payloads, launchpad, cores, flight number, date
- 3 Keep single-core, single-payload launches up to 13 Nov 2020
- 4 Call `/rockets`, `/launchpads`, `/payloads`, `/cores` by ID: booster version, site, lat/long, payload mass, orbit, outcome, grid fins, legs, reuse, block, serial
- 5 Keep `BoosterVersion != 'Falcon 1'`; fill missing `PayloadMass` with the mean → `dataset_part_1.csv`

Wikipedia web scraping

- 1 `requests.get()` the "List of Falcon 9 and Falcon Heavy launches" page (9 June 2021 snapshot)
- 2 `BeautifulSoup(response.text)` and `soup.find_all('table')`
- 3 Extract column names from the `<th>` cells of the first launch table
- 4 Loop over rows whose first cell is a flight number; clean date, booster, payload mass and landing with helper functions
- 5 `pd.DataFrame(launch_dict)` → `spacex_web_scraped.csv`

Notebooks: `1_spacex_data_collection_api.ipynb` · `2_spacex_web scraping.ipynb`

Data wrangling methodology

The raw Outcome column mixes landing success and landing type. It was turned into a binary training label.

Outcome	Meaning	Count	Class
True ASDS	Landed on a drone ship	41	1
True RTLS	Landed on a ground pad	14	1
True Ocean	Controlled ocean landing	5	1
None None	No landing attempted	19	0
False ASDS	Drone-ship landing failed	6	0
False Ocean	Ocean landing failed	2	0
None ASDS	Drone ship, no landing	2	0
False RTLS	Ground-pad landing failed	1	0

1 Missing values: only LandingPad (28.9%) is missing — kept as 'no pad used'

2 Launches per site: CCAFS SLC 40 55, KSC LC 39A 22, VAFB SLC 4E 13

3 bad_outcomes = every outcome not starting with 'True'

4 Class = 0 if outcome is bad, else 1 → mean success rate 0.667

5 One-hot encode Orbit, LaunchSite, LandingPad, Serial; cast to float64 (80 features)

Notebook: [3_spacex_data_wrangling.ipynb](#)

EDA methodology: visualization and SQL

Charts plotted (hue = Class)

Chart	Variables	Why
Scatter	Flight number vs. payload	Experience and payload together
Scatter	Flight number vs. launch site	Does each site improve over time?
Scatter	Payload vs. launch site	Site limits on payload
Bar	Success rate by orbit	Which orbits land best
Scatter	Flight number vs. orbit	Separate orbit from experience
Scatter	Payload vs. orbit	Heavy payloads per orbit
Line	Yearly success rate	Trend over time

SQL queries (SQLite, 101 records)

1. Unique launch sites — DISTINCT
2. Sites starting with 'CCA' — LIKE, LIMIT 5
3. Total NASA (CRS) payload — SUM
4. Average F9 v1.1 payload — AVG
5. First ground-pad success — MIN(Date)
6. Drone-ship success, 4-6 t payload — AND filters
7. Mission outcomes — GROUP BY, COUNT
8. Max-payload boosters — subquery + MAX
9. 2015 drone-ship failures — substr, CASE
10. Rank outcomes 2010-2017 — RANK() OVER

Notebooks: 5_spacex_eda_dataviz.ipynb · 4_spacex_eda_sql.ipynb

Interactive visual analytics methodology

Folium map

Circle + Marker (DivIcon)

One circle and label per launch site — where the 4 sites are

MarkerCluster + coloured Marker

Every launch: green if landed, red if not — success rate per site at a glance

MousePosition

Read lat/long under the cursor to pick coastline, rail, highway and city points

PolyLine + distance Marker

Lines from CCAFS SLC-40 to each feature, labelled with the haversine distance

Plotly Dash dashboard

Launch-site dropdown

All Sites or one site; drives both charts

Pie chart

All Sites: successes per site. One site: success vs. failure

Payload range slider

0-10,000 kg in 1,000 kg steps to filter launches

Scatter chart

Payload vs. outcome, coloured by booster version category

Files: 6_spacex_launch_site_folium.ipynb • 7_spacex_dash_app.py (two callbacks keep the charts in sync with the inputs)

Predictive analysis (classification) methodology

1 `Y = data['Class'].to_numpy(); X = 80 one-hot features (dataset_part_3)`

2 Standardize X with StandardScaler

3 `train_test_split(test_size=0.2, random_state=2) → 72 train / 18 test`

4 GridSearchCV(cv=10) for Logistic Regression, SVM, Decision Tree, KNN

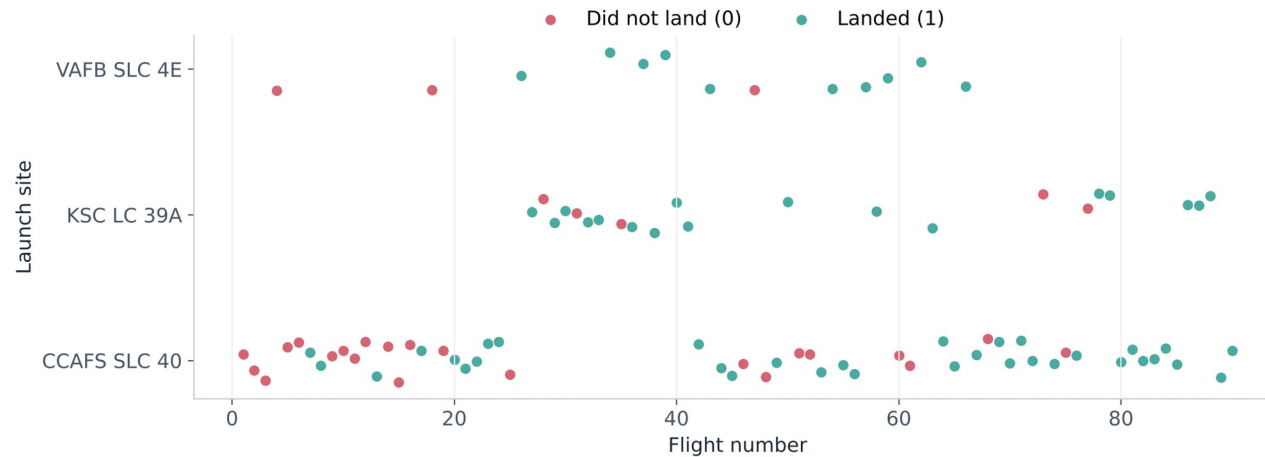
5 Score each best model on the test set and plot confusion matrices

6 Pick the best model by validation accuracy (test scores tie)

Model	Hyperparameters searched
Logistic Regression	$C \in \{0.01, 0.1, 1\}$, l2, lbfgs
SVM	kernel, C, gamma (logspace -3...3)
Decision Tree	criterion, splitter, max_depth 2-18, max_features, min_samples
KNN	n_neighbors 1-10, algorithm, $p \in \{1, 2\}$

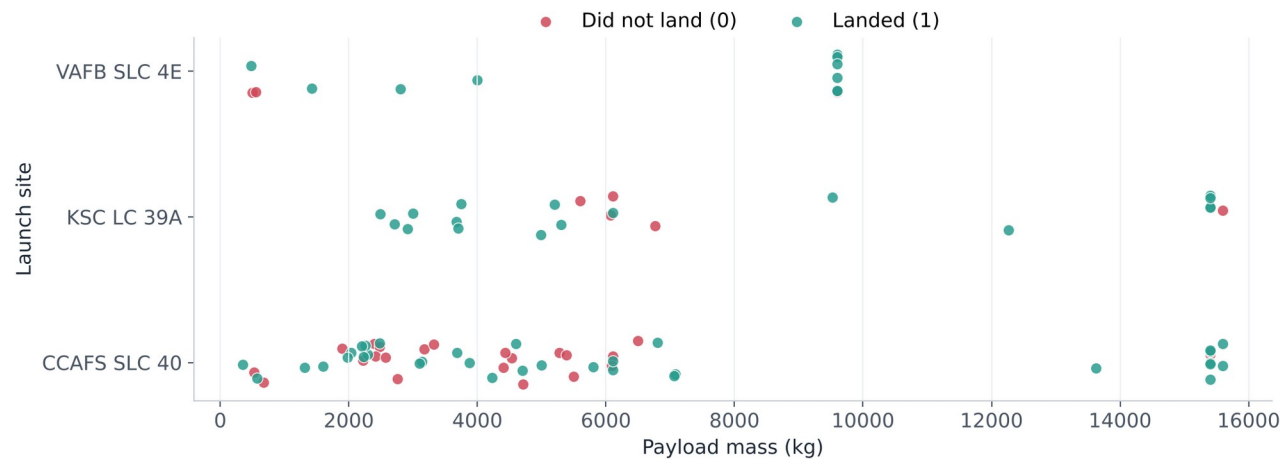
Notebook: [8_spacex_machine_learning_prediction.ipynb](#)

Flight number and payload vs. launch site



Flight number vs. launch site

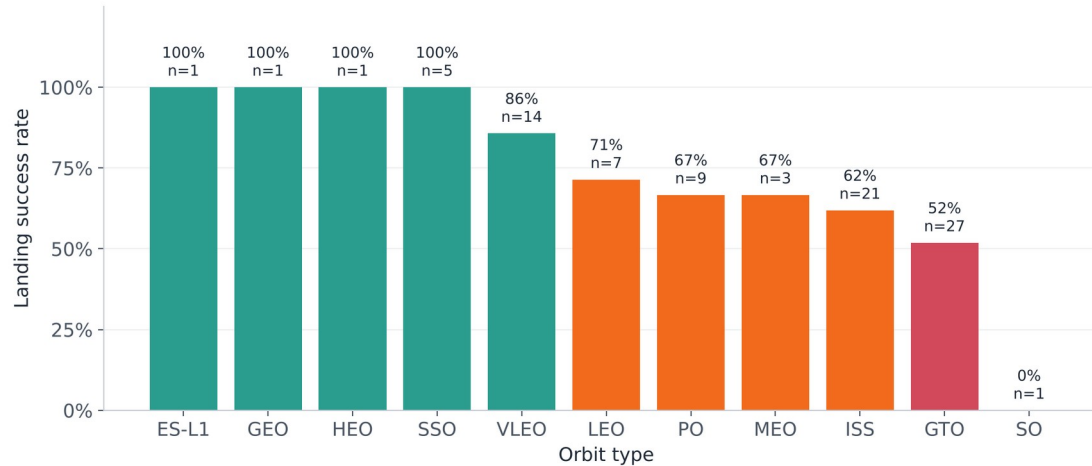
- CCAFS SLC 40 carries 55 of 90 launches, including nearly all early failures
- KSC LC 39A was used from flight ~27, when landings were more reliable
- At every site, failures fade as flight number grows



Payload vs. launch site

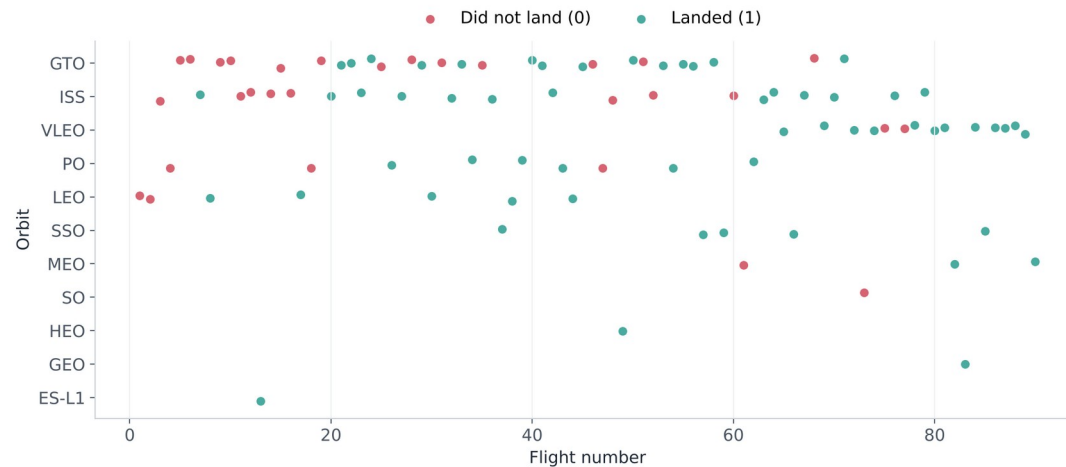
- VAFB SLC 4E launched no payload above 10,000 kg
- Heavy payloads (> 10,000 kg) flew from Florida and mostly landed
- Light and medium payloads show both outcomes at every site

Orbit type: success rate and flight number



Success rate by orbit type

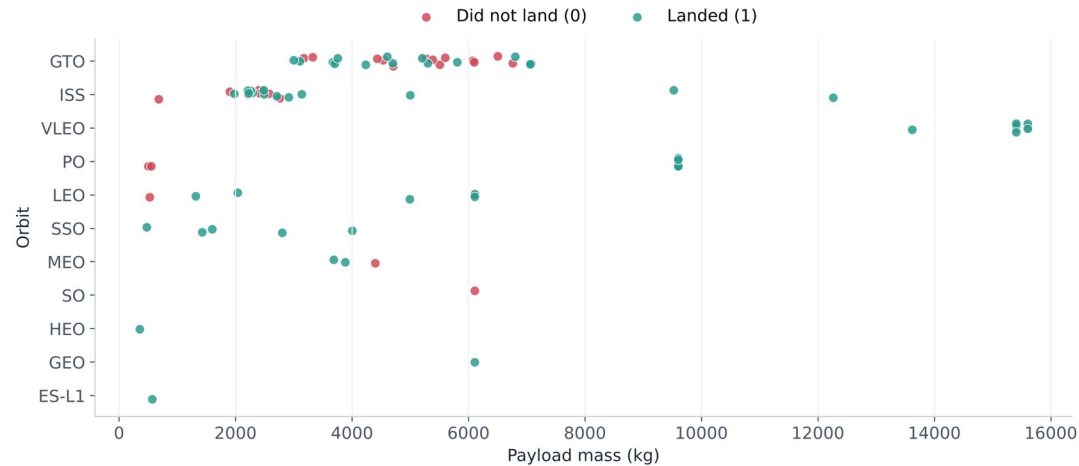
- ES-L1, GEO, HEO, SSO: 100% (only 1-5 launches each)
- VLEO 86%, LEO 71%
- GTO is the weakest frequent orbit: 52% of 27 launches



Flight number vs. orbit type

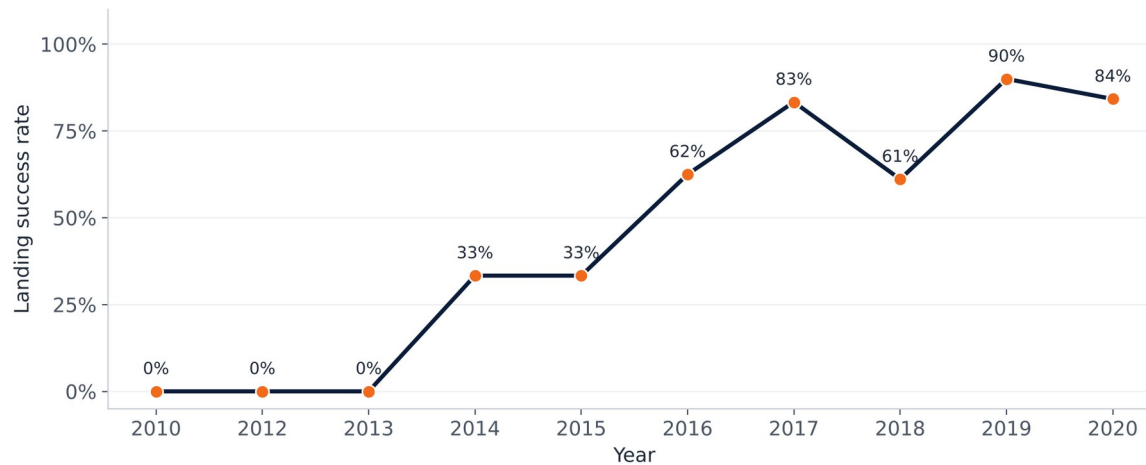
- LEO: success grows with flight number
- GTO: no clear relationship — failures continue throughout
- VLEO launches are recent and mostly successful

Payload vs. orbit and the yearly trend



Payload vs. orbit type

- With heavy payloads, Polar, LEO and ISS land more often
- GTO mixes landed and failed boosters across 3,000-7,000 kg
- VLEO payloads (~15,600 kg) mostly land



Launch success yearly trend

- 0% from 2010 to 2013; 33% in 2014-2015
- Climb to 63% (2016) and 83% (2017)
- Dip in 2018 (61%), then 90% in 2019 and 84% in 2020

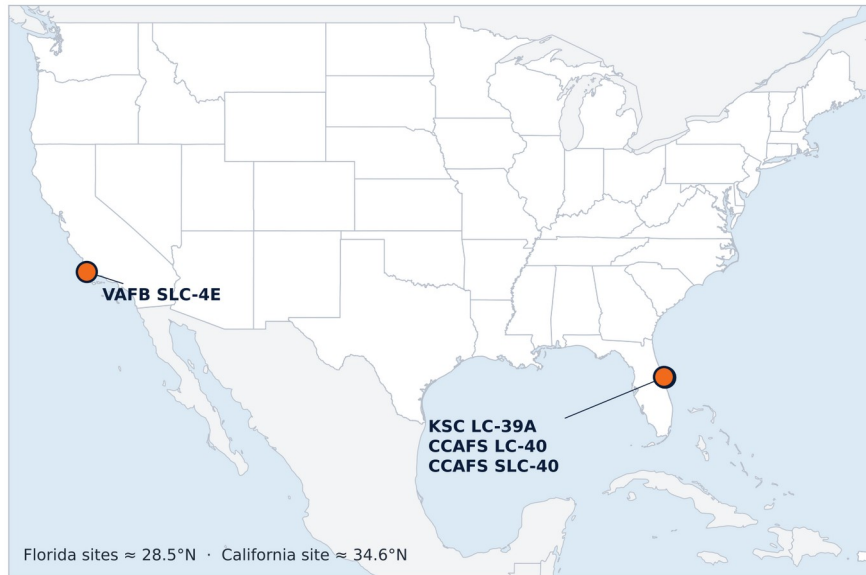
SQL results (tasks 1-5)

Task	SQL query	Result and explanation
1. Unique launch sites	<pre>SELECT DISTINCT Launch_Site FROM SPACEXTABLE;</pre>	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40 — four sites; LC-40 and SLC-40 are the same Cape Canaveral pad renamed in 2016.
2. Sites beginning with 'CCA'	<pre>SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;</pre>	5 early F9 v1.0 flights from CCAFS LC-40 (2010-06-04 to 2013-03-01), payloads 0-677 kg for SpaceX and NASA; landings failed (parachute) or not attempted.
3. Total payload, NASA (CRS)	<pre>SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';</pre>	45,596 kg carried for NASA's Commercial Resupply Services missions to the ISS.
4. Average payload, F9 v1.1	<pre>SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';</pre>	2,928.4 kg. Including every v1.1 serial (LIKE 'F9 v1.1%') gives about 2,534.7 kg.
5. First ground-pad landing	<pre>SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';</pre>	2015-12-22 — the first successful RTLS landing, almost 5 years after the first Falcon 9 flight.

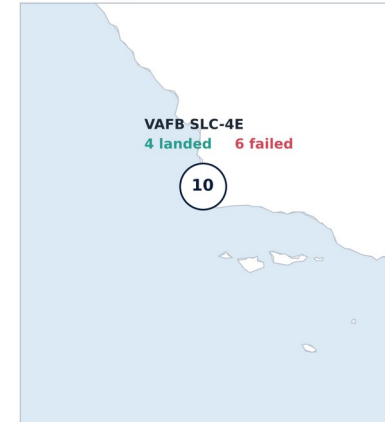
SQL results (tasks 6-10)

Task	SQL query	Result and explanation
6. Drone-ship success, 4,000-6,000 kg	<pre>SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome='Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;</pre>	F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2 — all Full Thrust boosters, two already on their second flight.
7. Mission outcomes	<pre>SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE GROUP BY Mission_Outcome;</pre>	Success 98 · Success (trailing space) 1 · Success (payload status unclear) 1 · Failure (in flight) 1 → 100 of 101 missions succeeded; the uncertainty is the landing.
8. Max-payload boosters	<pre>SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);</pre>	12 Block 5 boosters (15,600 kg): B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, B1049.7.
9. 2015 drone-ship failures	<pre>SELECT substr(Date,6,2) AS month, ... WHERE substr(Date,0,5)='2015' AND Landing_Outcome='Failure (drone ship)';</pre>	January: F9 v1.1 B1012, CCAFS LC-40 · April: F9 v1.1 B1015, CCAFS LC-40 (month names added with CASE).
10. Rank outcomes 2010-06-04 to 2017-03-20	<pre>SELECT Landing_Outcome, COUNT(*) n, RANK() OVER (ORDER BY COUNT(*) DESC) FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome;</pre>	No attempt 10 · Success (drone ship) 5 · Failure (drone ship) 5 · Success (ground pad) 3 · Controlled (ocean) 3 · Uncontrolled (ocean) 2 · Failure (parachute) 2 · Precluded (drone ship) 1.

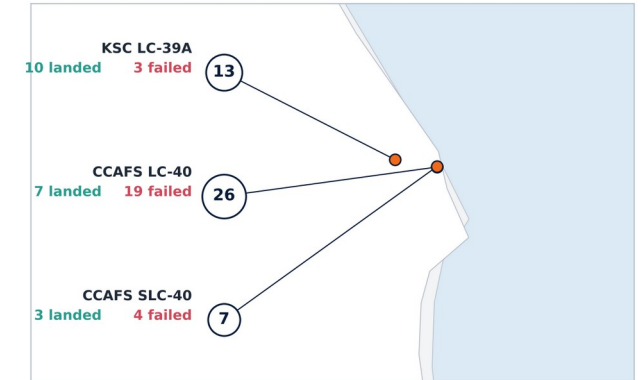
Folium: launch sites and launch outcomes



California



Florida (Cape Canaveral)

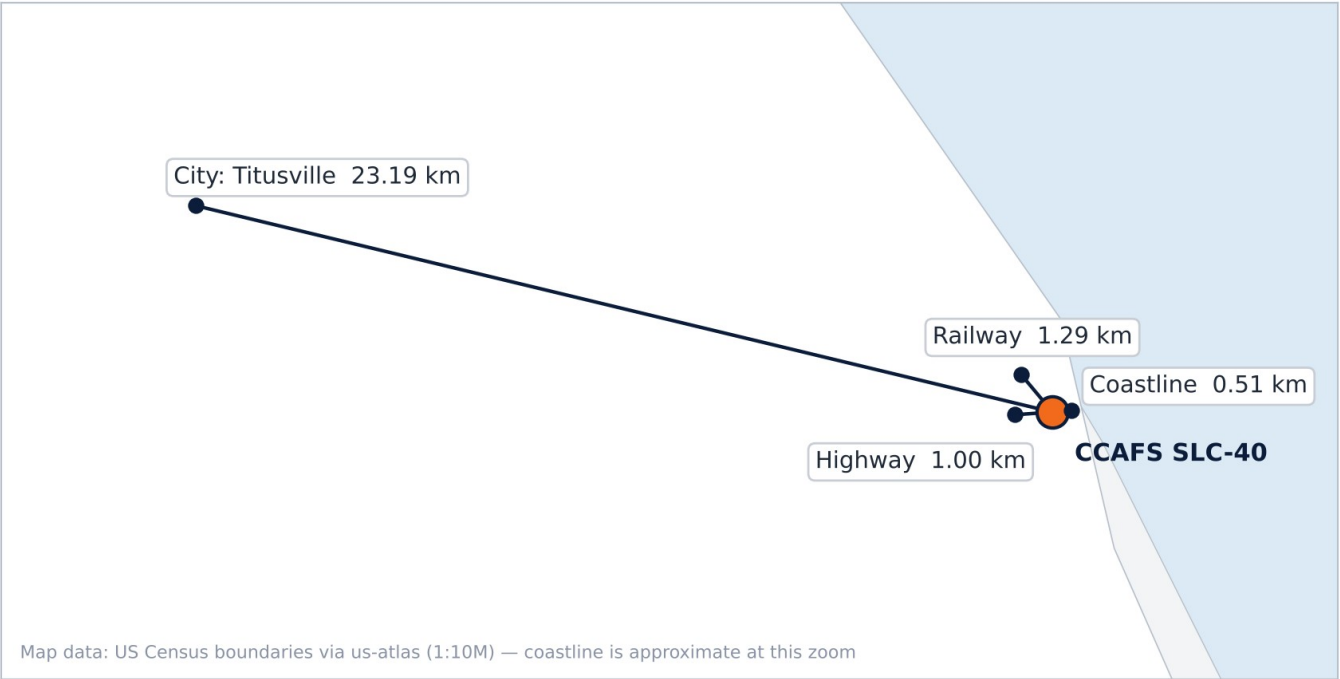


- 3 sites at Cape Canaveral, Florida (≈ 28.5°N), 1 at Vandenberg, California (≈ 34.6°N)
- All sites are on the coast and as far south as the mainland US allows (Earth's rotation helps)

- Green = landed, red = failed, clustered per site
- KSC LC-39A: 10 of 13 landed (77%) — best rate
- CCAFS LC-40: 7 of 26 (27%); VAFB SLC-4E: 4 of 10; CCAFS SLC-40: 3 of 7

Insight The busiest site has the lowest rate because it carried the early learning curve.

Folium: distances from CCAFS SLC-40 to its proximities



Feature	Distance
Coastline	0.51 km
Highway	1.00 km
Railway	1.29 km
City (Titusville)	23.19 km

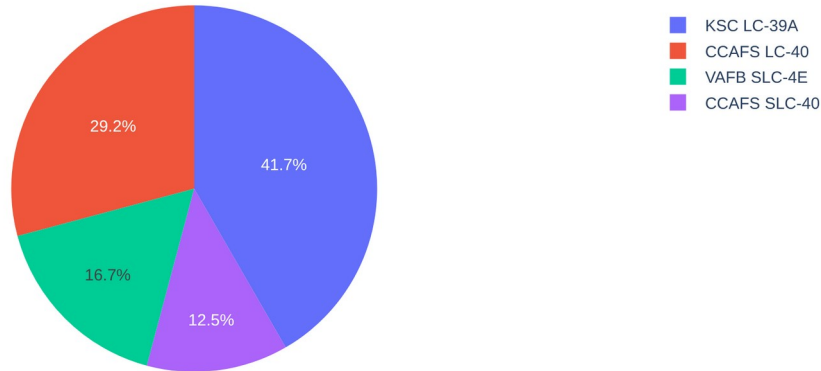
- Close to rail and highways to move heavy parts
- Right on the coastline
- ~23 km from the nearest city for safety

Insight Launch sites balance logistics (rail and roads within ~1 km) against safety (cities 20+ km away).

Dash: launch success by site

All sites — share of successful launches

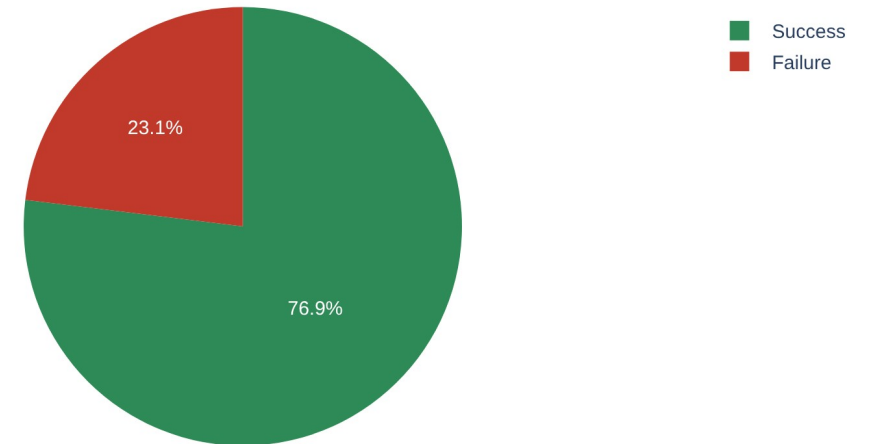
Total Successful Launches by Site



- KSC LC-39A gives 41.7% of all successful landings
- CCAFS LC-40 29.2%, VAFB SLC-4E 16.7%, CCAFS SLC-40 12.5%

KSC LC-39A — success vs. failure

Total Success vs. Failed Launches for site KSC LC-39A



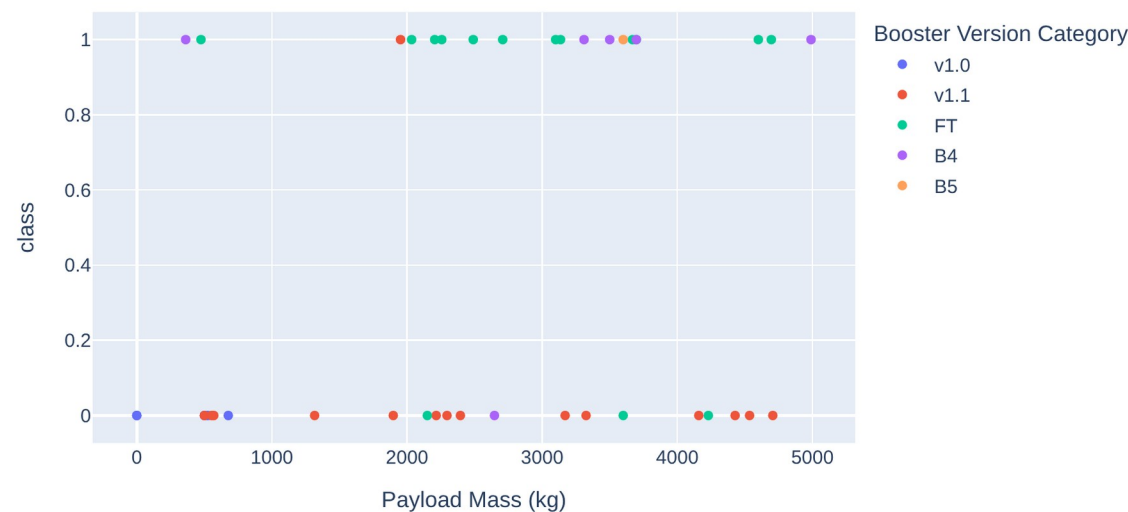
- KSC LC-39A has the highest success ratio: 76.9% (10) vs. 23.1% (3)
- Every other site is below 45%

Insight KSC LC-39A is both the biggest source of successes and the most reliable site.

Dash: payload vs. launch outcome for all sites

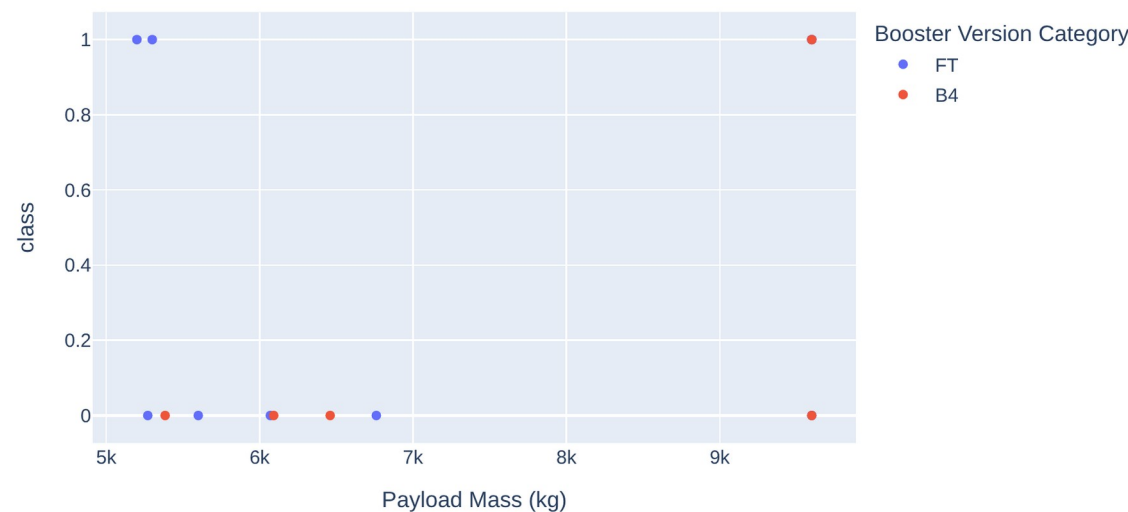
Payload 0 - 5,000 kg

Correlation between Payload and Success for all Sites



Payload 5,000 - 10,000 kg

Correlation between Payload and Success for all Sites

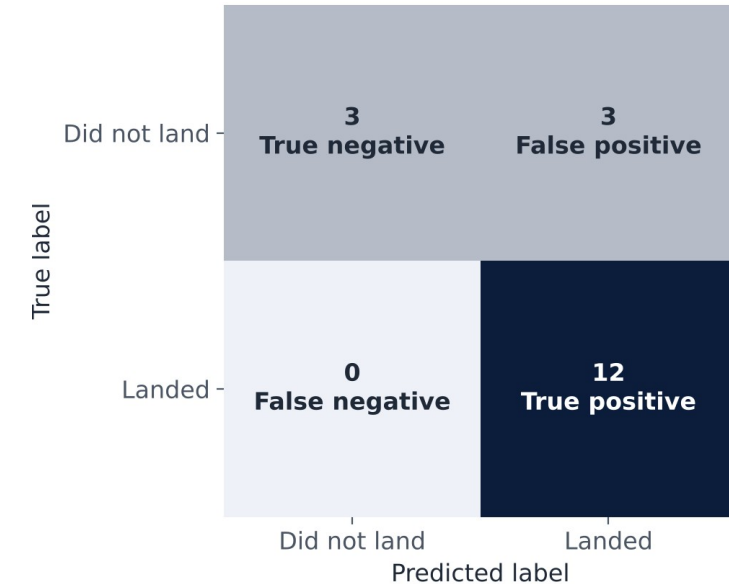
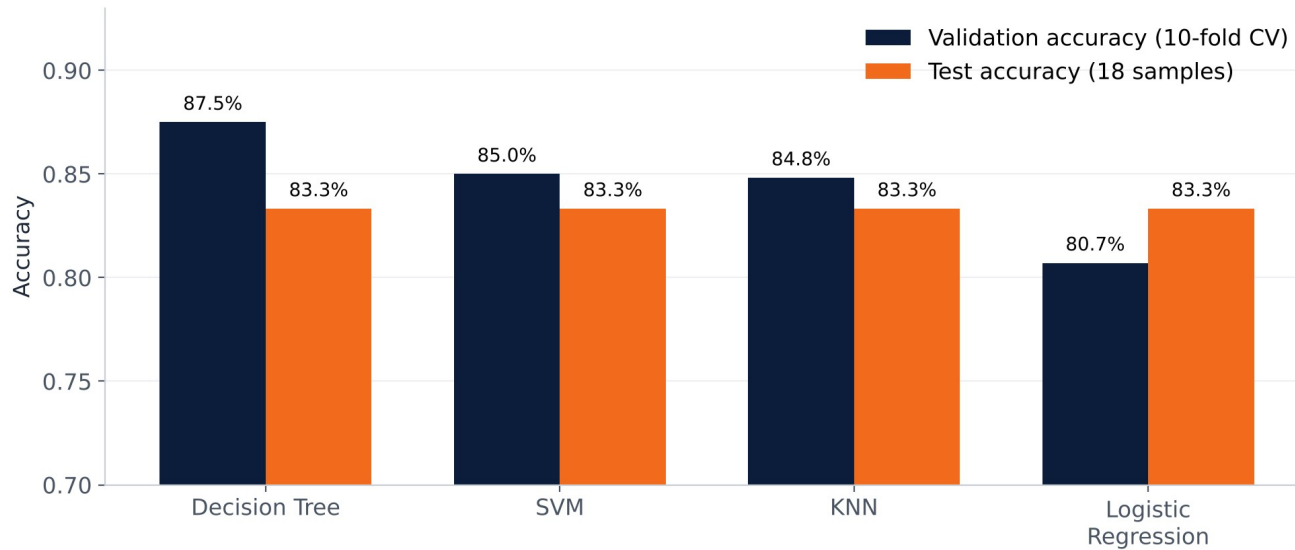


Payload band (kg)	0-2,000	2,000-4,000	4,000-6,000	6,000-10,000
Success rate	23%	62%	38%	33%
Launches	13	21	13	9

- 2,000-4,000 kg has the highest success rate; $\leq 5,000$ kg: 45% vs. $>5,000$ kg: 36%
- v1.0 and v1.1 boosters almost never landed; FT boosters landed 67% of the time

Insight Light payloads fail mostly because they flew on early v1.0/v1.1 boosters, not because they are light.

Classification results: accuracy and confusion matrix

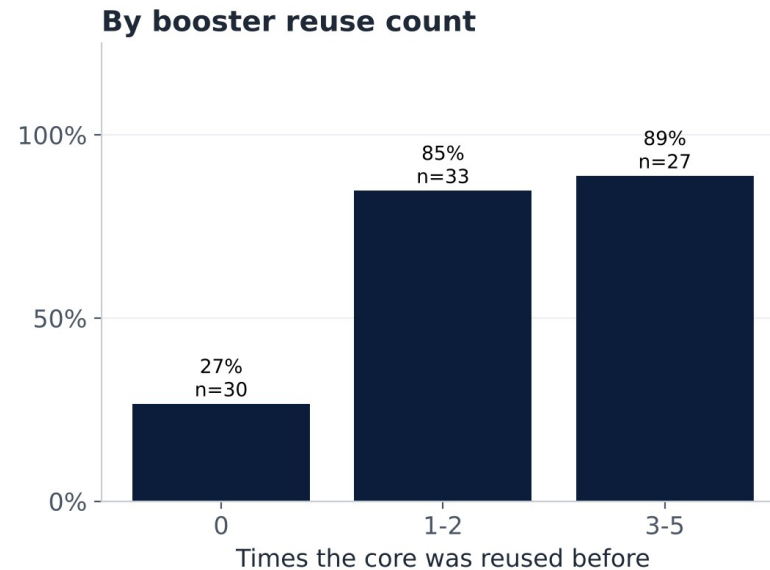
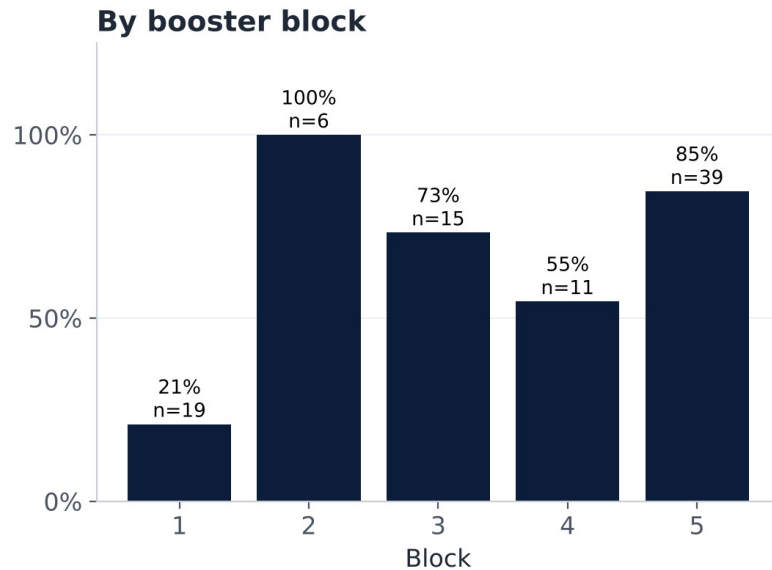


Model (best params)	CV acc.	Test acc.
Decision Tree	87.5%	83.3%
SVM (sigmoid, C=1)	85.0%	83.3%
KNN (k=6, p=1)	84.8%	83.3%
Logistic Reg. (C=0.1)	80.7%	83.3%

- Best model: Decision Tree (highest validation accuracy); all four score 15/18 on the test set
- Confusion matrix (same for all models): 12 TP, 3 TN, 3 FP, 0 FN
- Recall for 'landed' 100%, precision 80% — errors are false positives only

Insight With only 18 test samples one prediction moves accuracy by 5.6 points, so cross-validation decides the winner.

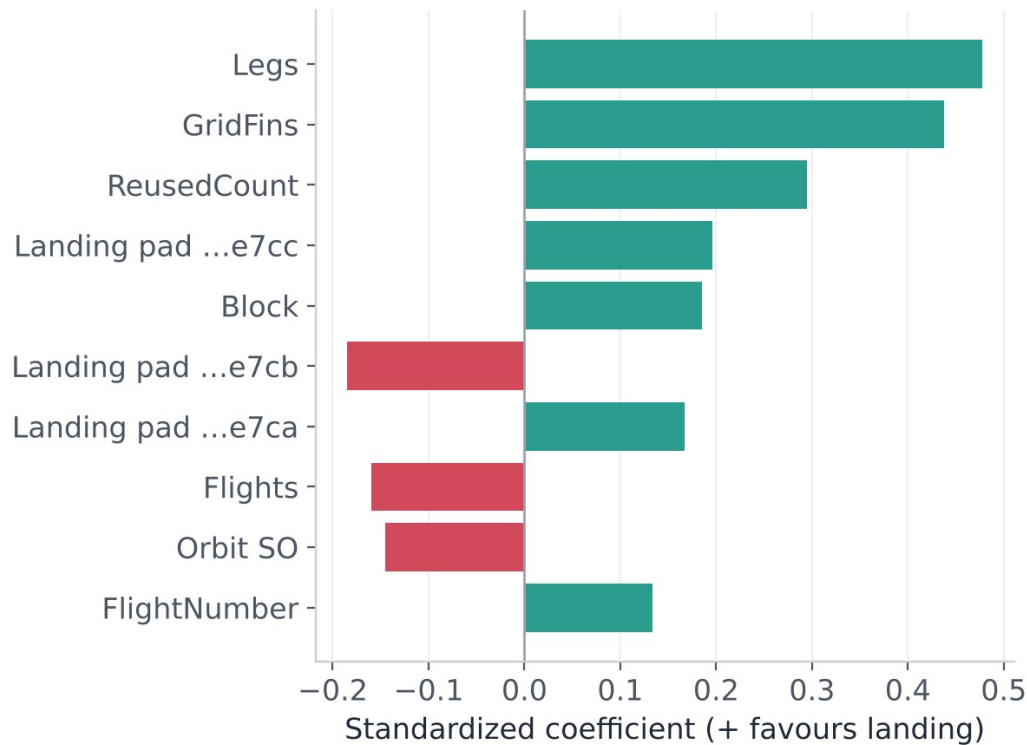
Innovative insight: reused, newer boosters land far more often



- Cores flying for the first time landed only 27% of the time (30 launches)
- Cores reused 1-2 times landed 85%; 3-5 times, 89%
- Block 5 boosters: 85% over 39 launches — the most reliable large group

Insight A booster that has landed before is very likely to land again — reuse history is a strong, cheap-to-collect predictor.

Innovative insight: landing drivers and bidding strategy



Top drivers (logistic regression)

Landing legs, grid fins and reuse count have the largest positive weights; block and flight number also help. These are hardware-readiness signals.

GTO is SpaceX's weak spot

GTO is the most frequent orbit (27 launches) but only 52% landed. When the first stage is lost, SpaceX's cost advantage shrinks.

Bidding strategy for Space Y

Target heavy GTO missions, where SpaceX is least likely to recover the booster.

Insight Serial-number dummies were excluded from the chart: they identify individual boosters and do not generalize.

Conclusion

1

Success is a learning curve

Landing rate grew from 0% (2010-13) to 90% in 2019; flight number, reuse and booster block matter most.

2

Orbit matters

ES-L1, GEO, HEO, SSO landed every time (small samples); GTO is the hardest common orbit at 52%.

3

KSC LC-39A is the best site

77% landing rate and 41.7% of all successes. All sites are coastal, close to rail and roads, and far from cities.

4

Landings are predictable

Tuned models reach 83.3% test accuracy; the Decision Tree is the best model (87.5% validation) and never misses a real landing.

5

Next steps

Add newer launches, weather and booster-history features, and tune the decision threshold to cut false positives.

Appendix

Project file	Content
1_spacex_data_collection_api.ipynb	SpaceX REST API collection and cleaning
2_spacex_webscraping.ipynb	Wikipedia scraping with BeautifulSoup
3_spacex_data_wrangling.ipynb	Outcome labelling (Class)
4_spacex_eda_sql.ipynb	10 SQL tasks on SQLite
5_spacex_eda_dataviz.ipynb	EDA charts and feature engineering
6_spacex_launch_site_folium.ipynb	Folium maps and distances
7_spacex_dash_app.py	Plotly Dash dashboard
8_spacex_machine_learning_prediction.ipynb	Model training and evaluation

Data notes

- Falcon 9 dataset: 90 launches, 2010-2020; model matrix 90×80 after one-hot encoding
- SQL dataset: 101 records (Spacex.csv)
- Dash / Folium dataset: 56 records
- Proximity distances come from points picked with MousePosition, so they are approximate
- Decision-tree tuning has randomness; re-runs can vary by a few points